

Research Paper

Pool boiling heat transfer of ammonia outside enhanced tubes with various fin structures

Jin Sub Kim^{a,b,*}, Wookyoung Kim^a, Hak Soo Kim^a, Young Kim^a^a Department of Thermal Energy Solutions, Korea Institute of Machinery and Materials, 156 Gajeongbuk-Ro, Yuseong-Gu, Daejeon 34103, Republic of Korea^b Department of Mechanical Engineering, University of Science of Technology, 217 Gajeong-Ro, Yuseong-Gu, Daejeon 34113, Republic of Korea

ARTICLE INFO

Keywords:

Pool boiling

Ammonia

Hysteresis

Enhanced tube

Enhancement factor

ABSTRACT

Pool boiling heat transfer of ammonia on a smooth tube and three enhanced tubes was investigated at the saturation pressure ranging from 6.15 to 11.7 bar, which corresponds to the saturation temperature from 10 to 30 °C. The test tube with a length of 796 mm was immersed in the liquid pool of ammonia inside a cylindrical chamber having an internal diameter of 254.4 mm, and convection or pool boiling heat transfer outside the test tube was generated by flowing hot water through the tube. Three enhanced tubes of a low-finned tube (Low-fin), a low-finned tube with oblique cuttings (Turbo-C), and a finned tube having re-entrant cavities (Turbo-E) were prepared to examine the pool boiling enhancement on various finned tubes. Boiling hysteresis on the smooth tube was observed between when the heat flux increases and when the heat flux decreases, which resulted in the deviation of the heat transfer coefficients from the previous correlations when the heat flux decreases. A suitable correlation was suggested to compensate for the deviation of the heat transfer coefficients in case of decreasing the heat flux by modifying the Stephan-Abdelsalam correlation. Three enhanced tubes exhibited considerable enhancement in the heat transfer coefficients compared with the smooth tube, and boiling performance on the Turbo-E tube having re-entrant cavities was highly affected by the variation of the bubble diameter according to the saturation pressure. Enhancement factors of the enhanced tubes were obtained from the ratio of thermal conductance on the enhanced tube to that on the smooth tube.

1. Introduction

As the impact of climate change on our environment becomes severe, technologies to alleviate global warming are getting more attention recently. Energy efficiency and carbon neutrality are the most urgent requirements for energy devices such as heat exchangers and heat pumps. In addition, industrial use of conventional fossil fuels such as coal and natural gas is being restricted to mitigate greenhouse gas emissions [1]. The use of synthetic refrigerants for HVAC (Heating, ventilation, and air-conditioning) systems is also limited because they generally have not a little global warming potential. Instead, the utilization of natural working fluids such as ammonia and water is recommended in the energy systems because they are free from carbon emissions. Ammonia has been widely used in the production of agricultural fertilizer, but limitedly utilized in industrial applications because of its flammable nature and toxicity. However, owing to its superior heat transfer capability and its distinct role as a hydrogen carrier, ammonia is regarded as a promising alternative to fossil fuels for

various industrial applications such as industrial heat pumps, waste heat recovery, and power plants.

Understanding the boiling phenomena of ammonia is essential to developing efficient energy systems because boiling is involved in the operation of most energy devices such as heat exchangers, heat pumps, and energy storage units. As a result, boiling heat transfer of ammonia has been investigated by many previous researchers concerning boiling performance, bubble behavior, and enhancement techniques. Zürcher et al. [2] investigated flow boiling heat transfer of ammonia inside a horizontal stainless steel tube at the saturation temperature of 4 °C and revealed that their flow boiling model gave fairly good predictions for the local heat transfer coefficients. Arima et al. [3] experimented forced convection boiling of ammonia in a vertical flat plate with a width of 2 mm and showed that local boiling heat transfer coefficients increased with increasing vapor quality. Maqbool et al. [4] also investigated flow boiling heat transfer of ammonia in two vertical smooth tubes having different internal diameters of 1.7 and 1.224 mm at three saturation temperatures of 23, 33, and 43 °C, and found that the heat transfer coefficient (HTC) increases with decrease in internal diameter. Some

* Corresponding author.

E-mail address: jskim129@kimm.re.kr (J.S. Kim).<https://doi.org/10.1016/j.applthermaleng.2024.122986>

Received 4 October 2023; Received in revised form 6 February 2024; Accepted 16 March 2024

Available online 18 March 2024

1359-4311/© 2024 Elsevier Ltd. All rights reserved.

Nomenclature

A	area (m ²)
A_t	total surface area including both base and finned area (m ²)
Bo_D	Bond number
C_{sf}	fluid-surface interaction constant for Rohsenow correlation
c_p	specific heat (J/kg K)
D	diameter (m)
D_b	bubble departure diameter (m)
D_f	outmost diameter including fins (m)
f	friction factor
g	gravitational acceleration (m ² /s)
h_b	boiling heat transfer coefficient (W/m ² K)
h_i	convective heat transfer coefficient in pipe (W/m ² K)
h_f	fin height (m)
h_{fg}	heat of vaporization (J/kg)
Ja^*	Jakob number
k	thermal conductivity (W/mK)
L	tube length (m)
$\dot{m}$	mass flow rate (kg/s)
Nu_D	Nusselt number
P	pressure (Pa)
P_c	critical pressure (Pa)
P_r	reduced pressure, = P/P_c
Pr	Prandtl number
Q	heat transfer rate (W)
q''	heat flux (W/m ²)

R_a	arithmetic average roughness (m)
R_{wall}	conduction thermal resistance through the pipe wall (K/W)
Re_D	Reynolds number
t_w	tube thickness (m)
T	temperature (K)
T_{in}	inlet temperature (K)
T_{out}	outlet temperature (K)
T_∞	pool temperature (K)
t_w	tube wall thickness (m)
U	overall heat transfer coefficient (W/m ² K)

Greek letters

ΔT_w	wall superheat (K)
ΔT_{lm}	log mean temperature difference (K)
β	contact angle (°)
η_o	overall fin efficiency
ρ	density (kg/m ³)
σ	surface energy (N/m)

Subscripts

D	diameter
i	inner
l	liquid
o	outer
sat	saturation
v	vapor
w	wall

researchers made efforts to reveal the heat transfer phenomena of ammonia spray cooling outside evaporator tubes. Zeng et al. [5] studied spray evaporation heat transfer of ammonia onto a horizontal triangular-pitch plain-tube bundle and found out that the spray evaporation coefficient can be as much as 65 % higher than the pool boiling coefficient. Yang et al. [6] examined the enhancement of spray cooling performance on the microcavity surfaces and observed that the microcavity surfaces showed higher HTC than the flat surface at high surface superheats because of the dominance of the nucleate boiling. The effect of surface modification on the spray cooling performance was investigated by Wang et al. [7] by using three modified surfaces – electrochemically treated, microporous coated, and hybrid surface combining macro fins and microporous coating. They found that the hybrid surface showed the best heat transfer enhancement owing to a larger surface area and higher boiling nucleation sites. In addition to flow boiling and spray cooling, pool boiling heat transfer of ammonia has also been studied by many researchers because pool boiling is a primary heat transfer mechanism in an evaporator of various energy systems such as chemical plants, distillation columns, absorption(adsorption) heat pumps, and turbo heat pumps. Inoue et al. [8] and Arima et al. [9] investigated pool boiling heat transfer of a binary mixture of ammonia/water on a horizontal heated wire and a horizontal heated surface, respectively, and found out the HTC in the mixture are smaller than those in both single component substances. Zheng et al. [10] examined the effect of miscible lubricant oil in ammonia on the pool boiling HTC and developed a correlation for the HTCs including the effect of the lubricant oil. Abbas et al. [11] presented a review of correlations for pool boiling of ammonia outside a single tube and bundles and suggested a careful application of the existing correlations about the pool boiling of ammonia because of a large disparity between various studies. Chyu et al. [12] conducted flooded boiling tests of ammonia/lubricant mixture on a bundle of horizontal enhanced tubes and found out that the HTC increased from the bottom row to the top row. Fernández-Seara et al. [13] compared pool boiling and spray evaporation heat transfer of

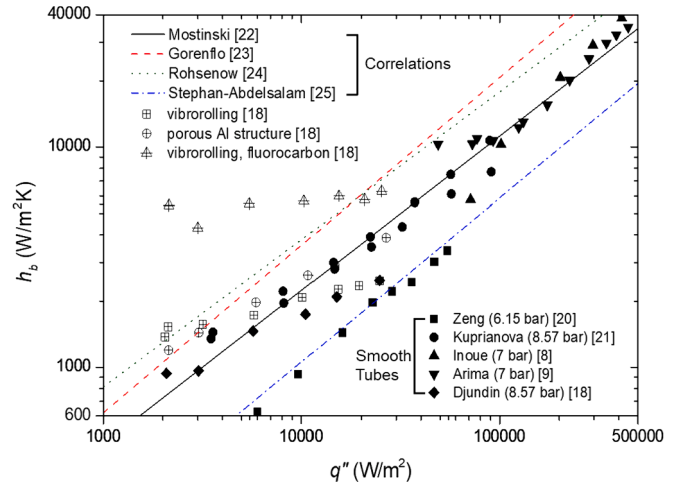


Fig. 1. Previous pool boiling heat transfer data of ammonia on smooth and enhanced tubes [8,9,18,20,21], along with representative correlations [22–25] (modified from Spindler [15]).

ammonia on a smooth tube, and observed that direct impingement on the tube surface enhanced the evaporation HTCs by 38.7 % on average. They also performed pool boiling enhancement tests on an integral-finned tube to determine the enhancement factor compared with the heat transfer performance on a plain tube [14]. Spindler [15] provided a comprehensive overview of the pool boiling heat transfer behavior of ammonia, and Ayub [16] also presented an overview of the boiling heat transfer enhancement of ammonia for the development of compact heat exchangers.

Fig. 1 shows some experimental data from previous pool boiling studies of ammonia on both smooth and enhanced surfaces at saturation

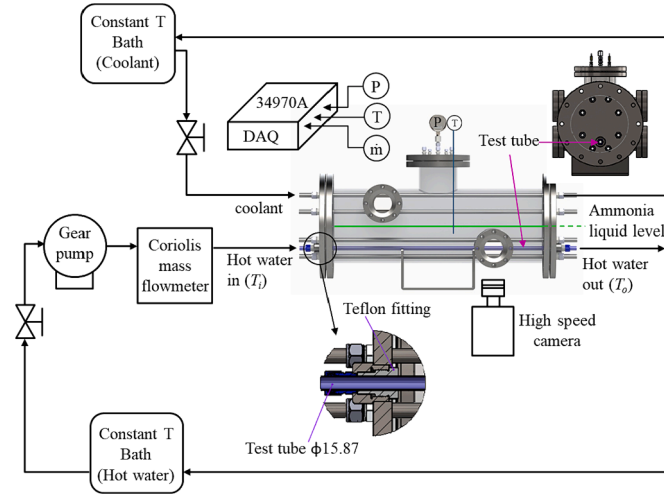


Fig. 2. Experimental apparatus for pool boiling heat transfer outside a single tube in ammonia.

pressures ranging from 6.15 to 8.57 bar, along with the representative pool boiling correlations. It is noticeable that there is a large discrepancy among pool boiling HTC's on the smooth surfaces measured by various researchers. Pool boiling characteristics of ammonia even on a smooth surface are not clearly revealed and more investigation is needed to clarify why there is a large difference in the experimental data. Moreover, there are limited numbers of experimental data about pool boiling heat transfer on the enhanced surface such as low-finned tubes except for the rigorous studies conducted by a group of Russian scientists [17,18]. As pointed out in the review paper [19], new experimental data on pool boiling of ammonia are insufficient to derive a more precise correlation on both smooth and enhanced surfaces. For this reason, the present study aims to investigate the pool boiling heat transfer of ammonia on the smooth tube and reveal the difference in the HTC's between when the heat flux increases and when the heat flux decreases. Based on the measured data, the relevant correlation for the HTC when decreasing the heat flux is suggested. Pool boiling heat transfer performance on three enhanced tubes is compared with that on the smooth tube, and the enhancement factors for the enhanced tubes are derived.

2. Experimental setup

2.1. Experimental apparatus and tube geometry

Fig. 2 shows the pool boiling test apparatus for measuring boiling performance outside a single tube in a flooded-type evaporator of ammonia. The test tube with an outer diameter of 15.87 mm (smooth tube) was located at the bottom center of a cylindrical test chamber made of stainless steel. The entire test tube with a length of 796 mm is submerged in an ammonia pool because almost half of the test chamber is filled with liquid ammonia. The test chamber has an internal dimension of $\phi 254.4$ mm (diameter) $\times$ 796 mm (length), and a chamber wall

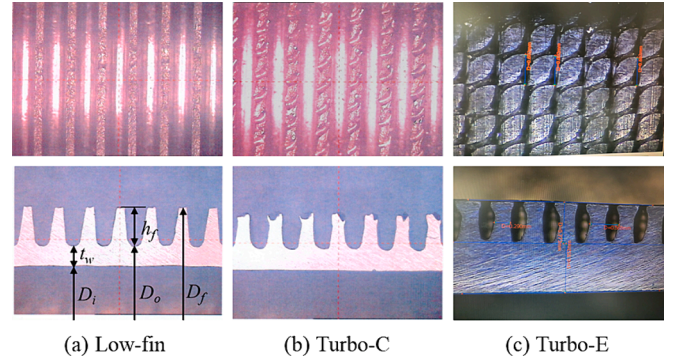


Fig. 3. Enlarged and cross-sectional images of three enhanced tubes as well as the definition of the important dimensions.

thickness is 6.5 mm. A Teflon fitting was inserted between the test tube and the chamber flange to minimize the heat loss transferred to the stainless steel flanges as shown in the enlarged image in Fig. 2. Wall temperature of the test tube was increased to the preset values by flowing hot water through the tube. A gear pump (ISMATEC, MCP-Z) drives the hot water through the test tube at a constant flow rate of 3.8 kg/min, which is measured by a Coriolis flowmeter (TRICOR, TCM 650). A constant temperature bath was employed to maintain the hot water temperature at the set value. Inlet and outlet temperatures of the hot water were measured by using two ultra-precise RTDs with 1/10 DIN accuracy (OMEGA, P-M-1/10-1/8), which has an accuracy of ± 0.03 °C at the temperature of 0 °C. There are eight stainless steel tubes circumferentially located in the test chamber, and coolant (ethylene glycol-water mixture) flows through the tubes to adjust the internal temperature of the chamber at the set value. Internal temperature and pressure of the test section were measured using a T-type thermocouple and an absolute pressure transducer (OMEGA, PX409), respectively. All measured data including temperature, pressure, and mass flow rate were collected into a data acquisition system (Keysight, 34970A), which was controlled by its software, Benchlink Data Logger. Bubble generation and departure behavior on the test tube were observed through a glass window installed at the side of the chamber, and a high-speed camera (Phantom, Miro 320S) was utilized to acquire the bubble images at a frame rate of 2000 fps (frames per second).

One smooth tube and three enhanced tubes were fabricated for the boiling performance tests and the specifications of four tubes are summarized in Table 1. Fig. 3 displays enlarged and cross-sectional images of three enhanced tubes as well as the definition of the important dimensions. The smooth tube made of stainless steel 316 has inner and outer diameters of 13.39 and 15.87 mm, respectively. The arithmetic average roughness, R_a , of the smooth tube was 0.296 mm, which was measured using a surface roughness tester (Mitutoyo, SJ-410). Since fin geometries on the enhanced tubes are formed by processing the smooth tube with a rolling machine, the diameter for the enhanced tube including the fins, D_f , is comparable to the outer diameter of the smooth tube. Besides, the material of each tube was also changed owing to the difficulties involved in forming the fin geometries. While Low-fin and

Table 1
Detailed dimensions of one smooth tube and three enhanced tubes.

	Material	D_i	D_o	t_w	D_f	h_f	Number of fins (#/inch)	Surface area per unit length (m^2/m)
Smooth	Stainless steel 316	13.39	15.87	1.24	—	—	—	0.04986
Low-fin	Stainless steel 439	11.81	13.07	0.63	15.73	1.33	26	0.1477 ($2.96 \times A_{t,smooth}$)
Turbo-C	Stainless steel 439	11.84	13.21	0.685	15.97	1.205	26	0.1264 ($2.54 \times A_{t,smooth}$)
Turbo-E	Carbon steel A179	12.24	14.16	0.96	15.68	0.76	44	0.1892 ($3.79 \times A_{t,smooth}$)

Turbo-C tubes were made of stainless steel 439, the Turbo-E tube with more complicated fin geometries was made of carbon steel A179, that is, more soft material. Low-fin and Turbo-C tubes have fins with a height of 1.33 and 1.205 mm, but the Turbo-E tube has fins with a height of 0.77 mm. However, the total finned surface area of the Turbo-E tube is larger than the other two enhanced tubes owing to a greater number of fins (44 fins per inch). It is noticeable that the Turbo-E tube has re-entrant type cavities with a large cavity and small cavity mouth which are known for facilitating bubble nucleation at smaller wall superheat [26].

2.2. Experimental procedure and data reduction

The experimental procedure and data reduction method are in detail described in the authors' previous paper [27] as well. A pool boiling experiment on a smooth tube and three enhanced tubes was conducted at three saturation temperatures of 10, 20, and 30 °C, which corresponds to the saturation pressure of 6.15, 8.57, and 11.7 bar, respectively. At first, the internal temperature of the chamber is adjusted to the wanting saturation temperature with the aid of coolant flow through the circumferential tubes as well as hot water flow through the test tube. At that time, both coolant and hot water are maintained at the identical saturation temperature. Once the internal temperature (pressure) reaches the wanting saturation temperature (pressure), the hot water temperature increases to the next setting value and the coolant temperature decreases to maintain the internal pool temperature (pressure) constant. Hot water flow increases wall temperature outside of the test tube, and this wall superheat brings about either convection heat transfer or nucleate boiling heat transfer. When the inlet and outlet temperatures of hot water reach the steady state, all data including the pool temperature, internal pressure, and inlet and outlet temperatures of hot water are stored and the hot water temperature increases to the next step (increasing sequence). Once the hot water temperature reaches the final set value, it is decreased to the lower value in the reverse order (decreasing sequence). When the hot water temperature arrives at the initial value, the entire test is terminated.

Heat transfer coefficient (HTC) outside the test tube was obtained from the following overall heat transfer coefficient, U , relation.

$$\frac{1}{UA} = \frac{1}{h_i A_i} + R_{wall} + \frac{1}{h_b A_o} \quad (1)$$

where h_i refers to the forced convection HTC inside the tube and h_b refers to the pool boiling (or natural convection) HTC outside the tube. R_{wall} indicates wall conduction thermal resistance through the tube wall expressed in the following relation.

$$R_{wall} = \frac{\ln(D_o/D_i)}{2\pi kL} \quad (2)$$

UA can be obtained by combining the energy balance of hot water flow and log-mean temperature difference (LMTD) relations as shown in Eq. (3) and (4).

$$Q = \dot{m}c_p(T_{in} - T_{out}) \quad (3)$$

$\dot{m}$ refers to the mass flow rate of hot water flowing through the test tube.

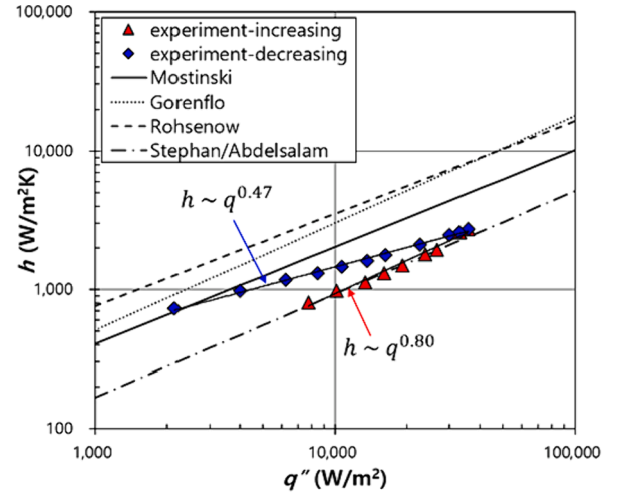
$$Q = UA\Delta T_{lm} \quad (4)$$

where ΔT_{lm} refers to the log-mean temperature difference defined in Eq. (5).

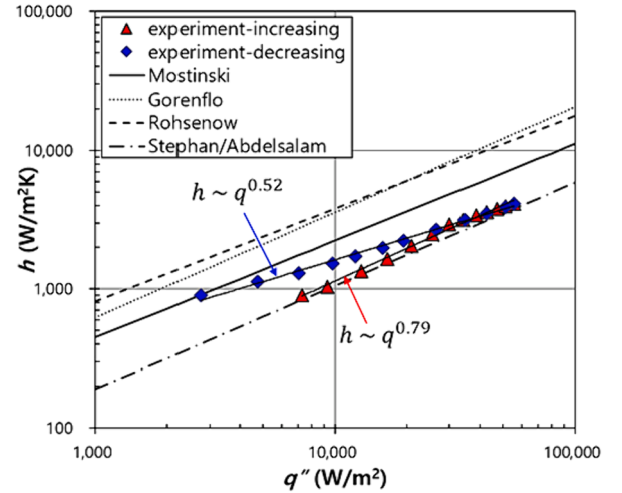
$$\Delta T_{lm} = \frac{\Delta T_{in} - \Delta T_{out}}{\ln(\Delta T_{in}/\Delta T_{out})} \quad (5)$$

$$\Delta T_{in} = T_{in} - T_{\infty}$$

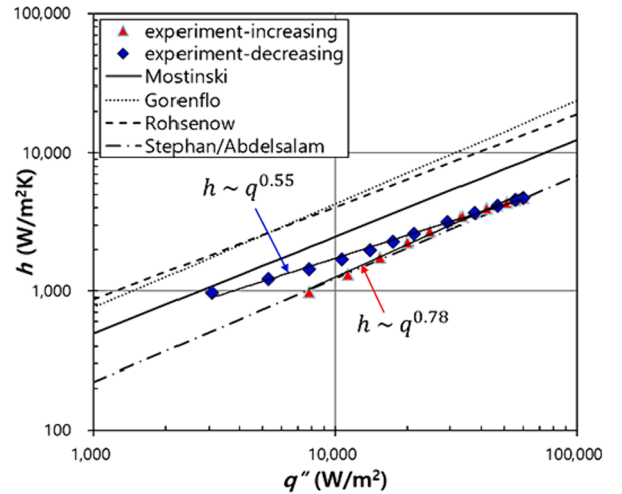
$$\Delta T_{out} = T_{out} - T_{\infty}$$



(a) $P_{sat} = 6.15$ bar ($T_{sat} = 10$ °C)



(b) $P_{sat} = 8.57$ bar ($T_{sat} = 20$ °C)



(c) $P_{sat} = 11.7$ bar ($T_{sat} = 30$ °C)

Fig. 4. Measured heat transfer coefficients at three saturation pressures in comparison with previous pool boiling correlations.

where T_∞ refers to the ambient pool temperature outside the tube, which is identical to the saturation temperature. The forced convection HTC inside the tube, h_i , was obtained from the Gnielinski correlation [28] regarding turbulent flow heat transfer inside a tube (Eq. (6)).

$$Nu_D = \frac{(f/8)(Re_D - 1000)Pr}{1 + 12.7(f/8)^{1/2}(Pr^{2/3} - 1)} \quad (6)$$

where $f = (0.790 \ln Re_D - 1.64)^{-2}$

where Re_D refers to the Reynolds number based on the inner diameter of the test tube and the Gnielinski correlation is valid over a large Reynolds number range from 3000 to 5×10^6 . In the present study, Reynolds number ranges from 5266 to 14,733 according to the variation of inlet and outlet water temperatures, even though the mass flow rate of hot water through the tube is controlled as 3.8 kg/s. Substituting R_{wall} , UA , and h_i obtained from Eq. (2) to Eq. (6) into Eq. (1), the remaining unknown, boiling HTC, h_b , can be obtained. The ratio of the thermal resistance of internal convection ($1/(h_i A_i)$) to the overall thermal resistance ($1/(UA)$) varies according to the heat flux and the tube geometry. For all the smooth and enhanced tubes tested, the ratio $[1/(h_i A_i)]/[1/(UA)]$ ranges from 0.255 to 0.697.

Uncertainties for main variables such as boiling HTC, heat transfer rate (Q), and wall superheat (ΔT) were calculated according to the guidance by Moffat [29]. Ultra-precise RTD sensors for measuring inlet and outlet temperatures have a 1/10 DIN accuracy class with a tolerance of $\pm 0.03^\circ\text{C}$ at the temperature of 0°C , and a T-type thermocouple for measuring ammonia pool temperature has an uncertainty of $\pm 0.5^\circ\text{C}$. The uncertainty of the mass flow rate measured by a Coriolis mass flowmeter was $\pm 0.1\%$ of the reading. The uncertainty of the internal heat transfer coefficient, h_i obtained from the Gnielinski correlation was estimated to be 2.66 % in the Reynolds number range tested with reference to previous studies [30,31] comparing experimentally measured data with the correlation. Through the error propagation of measured variables, the averaged expanded uncertainties for the boiling HTC, heat transfer rate (Q), and wall superheat (ΔT_w) were calculated as 18.0 %, 4.2 %, and 19.0 %, respectively. The averaged expanded uncertainty for the heat transfer enhancement factor, which is defined as the ratio of thermal conductance between smooth tube and enhanced tube, was estimated as 23.1 %.

3. Results and discussion

3.1. Boiling heat transfer correlation on the smooth tube

The HTC of ammonia on the smooth tube was measured at three saturation pressures of 6.15, 8.57, and 11.7 bar which corresponds to the saturation temperature of 10, 20, and 30°C . Fig. 4 shows the measured HTCs at three saturation pressures in comparison with previous pool boiling correlations. Four renowned correlations, Mostinski [22], Gorenflo [23], Rohsenow [24], and Stephan/Abdelsalam [25] were selected for the comparison. (Eq. (7–10))

$$h_b = 0.1011(P_c)^{0.69} q^{0.7} \left[1.8 \left(\frac{P}{P_c} \right)^{0.17} + 4 \left(\frac{P}{P_c} \right)^{1.2} + 10 \left(\frac{P}{P_c} \right)^{10} \right] \quad (7)$$

where P_c : critical pressure (bar)

$$h_b = h_0 \left(\frac{q}{q_0} \right)^{n(P_r)} F(P_r) \left(\frac{R_a}{R_{a0}} \right)^{0.133} \quad (8)$$

$$F(P_r) = 1.2P_r^{0.27} + \left(2.5 + \frac{1}{1 - P_r} \right) P_r$$

$$n(P_r) = 0.9 - 0.3P_r^{0.3} \text{ where } P_r: \text{reduced pressure } (P/P_c)$$

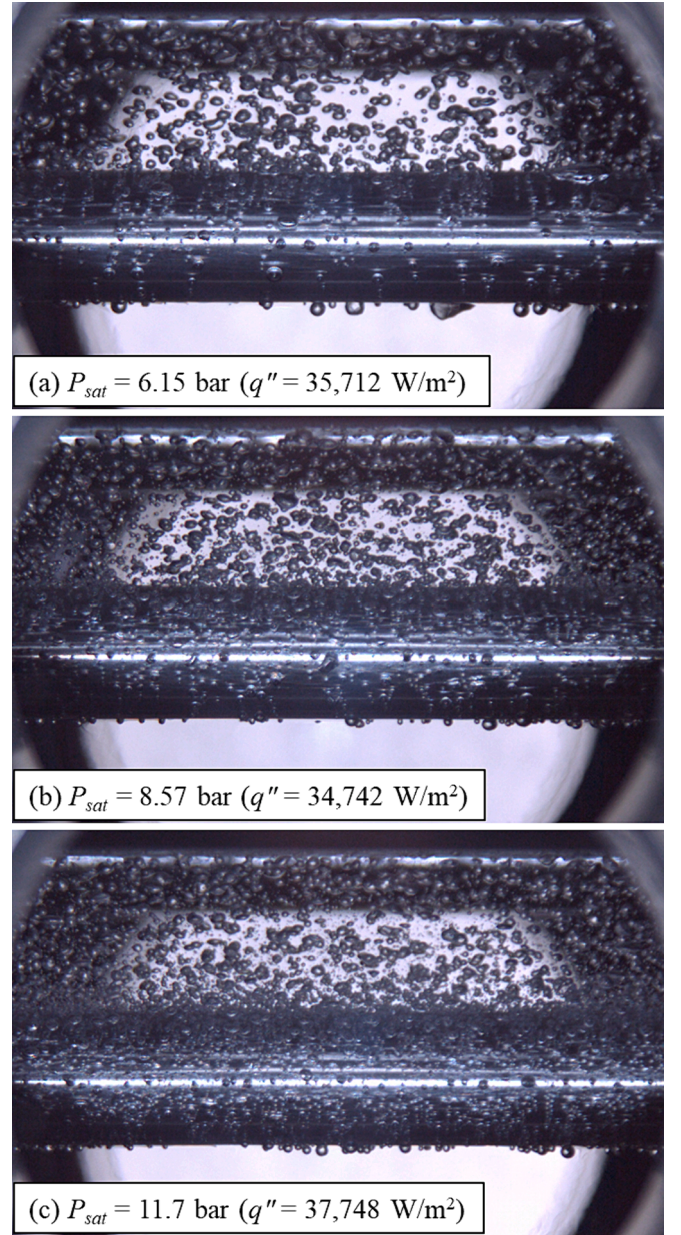


Fig. 5. Variation of bubble departure diameter and number of nucleation sites according to the saturation pressure ($q'' \sim 35,000 \text{ W/m}^2$).

$$h_b = \frac{1}{C_{sf}} \frac{C_{pl}}{h_{fg}^{0.67}} \left[\frac{1}{\mu_l} \sqrt{\frac{\sigma}{g(\rho_l - \rho_v)}} \right]^{-0.33} (q'')^{0.67} (Pr_l)^{-1.7} \quad (9)$$

where $C_{sf} = 0.013$ (first approximation)

$$\frac{h_b D_b}{k_l} = 207 \left(\frac{q D_b}{k_l T_{sat}} \right)^{0.745} \left(\frac{\rho_v}{\rho_l} \right)^{0.581} (Pr_l)^{0.533} \quad (10)$$

$$D_b = 0.0146 \beta \left[\frac{2\sigma}{g(\rho_l - \rho_v)} \right]^{1/2}$$

contact angle $\beta = 35^\circ$ for ammonia.

Above all, large hysteresis between increasing heat flux and decreasing heat flux can be seen in boiling heat transfer of ammonia. A higher HTC is obtained when the heat flux decreases than when the heat flux increases. For example, in the case of the saturation pressure of 6.15 bar ($T_{sat} = 10^\circ\text{C}$) the HTC when the heat flux decreases is $1457 \text{ W/m}^2 \text{ K}$ at the heat flux of $10,665 \text{ W/m}^2$, which is 49 % higher than the HTC

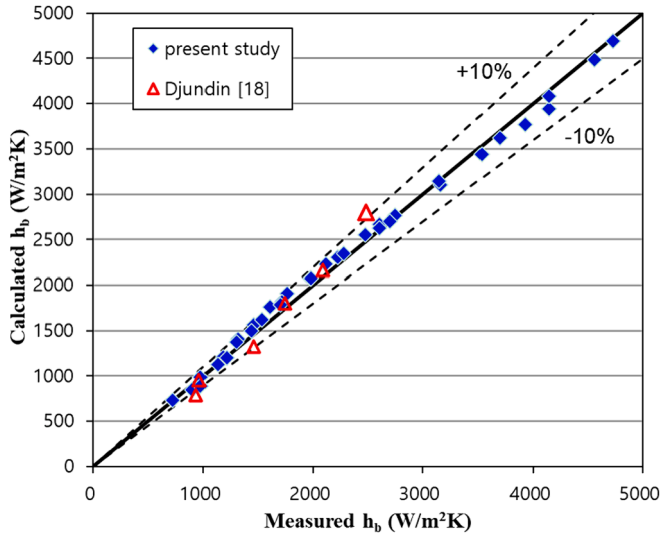
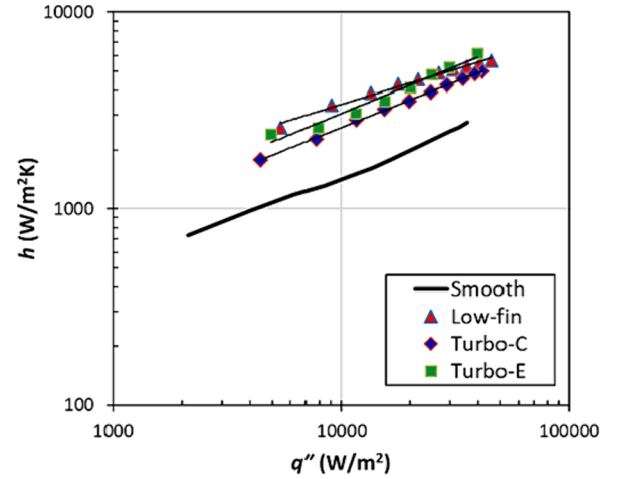


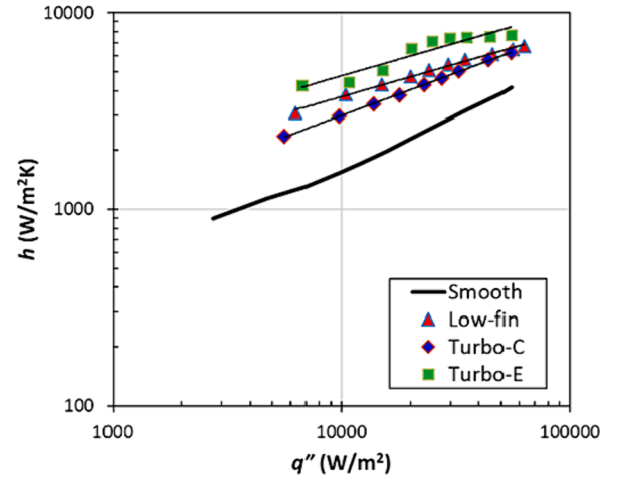
Fig. 6. Comparison between the measured HTC and the HTC calculated from Eq. (12) for the case of decreasing the heat flux.

(975 W/m² K) when the heat flux increases at a similar heat flux of 10,150 W/m². The large hysteresis is more prominent at the low heat flux region because it is related to the bubble nucleation mechanism on the wall cavity. Kim et al. [27] explained that larger wall superheat is required when a bubble grows from the small embryonic bubble inside the cavity (increasing the heat flux), but once the bubble is generated, small wall superheat is enough to generate successive bubbles because of the large bubble remaining from the departure of the previous bubble (decreasing the heat flux). It is reasonable that the hysteresis between increasing heat flux and decreasing heat flux is diminished with increasing the saturation pressure because the wall superheat required to generate bubbles decreases with the saturation pressure [27]. Bubble nucleation occurs at a smaller wall superheat as the saturation pressure increases, which is also responsible for the smaller bubble departure diameter at the higher saturation pressure. On the contrary, bubble departure frequency increases with the saturation pressure because the threshold (required wall superheat) for the bubble nucleation is lowered at the higher saturation pressure. Fig. 5 depicts photographs of the variation of the bubble departure diameter and the number of bubbles according to three saturation pressures at a similar heat flux of approximately 35,000 W/m². A much larger number of bubbles with a smaller diameter are generated on the smooth tube surface at the higher saturation pressure of 11.7 bar than those at the saturation pressure of 6.57 bar.

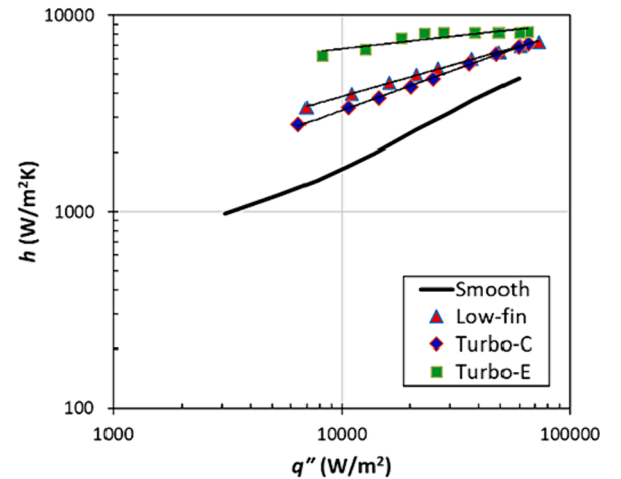
The HTCs in the case of increasing heat flux are well matched with the Stephan/Abdelsalam correlation among four previous correlations in Fig. 4. In contrast, the decreasing heat flux case does not match with any of the four previous correlations. The measured data in the case of decreasing heat flux are comparable to the Stephan/Abdelsalam correlation at the high heat flux region above 30,000 W/m², but as the heat flux decreases, the data deviates from the correlation and approaches the Mostinski correlation. The actual heat exchanger is more likely to follow the boiling curve of decreasing heat flux because there is continual up-and-down fluctuation in heat flux for a long operation time. Moreover, fully nucleate boiling can be attained by applying preheating for activating bubble nucleation cavities, if necessary. Therefore, it is most of all important to derive an appropriate correlation for the decreasing heat flux case in pool boiling of ammonia. As we already noticed the effect of saturation pressure on the boiling hysteresis, the empirical correlation for the case of decreasing heat flux was devised by incorporating the effect of saturation pressure into the modified form of the Stephan/Abdelsalam correlation (Eq. (11)).



(a) $P_{sat} = 6.15$ bar ($T_{sat} = 10$ °C)



(b) $P_{sat} = 8.57$ bar ($T_{sat} = 20$ °C)



(c) $P_{sat} = 11.7$ bar ($T_{sat} = 30$ °C)

Fig. 7. Boiling HTCs on three enhanced tubes of Low-fin, Turbo-C, and Turbo-E compared with those on the smooth tube (in case of decreasing the heat flux).

$$\frac{h_b D_b}{k_l} = C_1 \left(\frac{q'' D_b}{k_l T_{sat}} \right)^{n(P_r)} \left(\frac{\rho_v}{\rho_l} \right)^{0.581} (Pr_l)^{0.533} \quad (11)$$

$$D_b = 0.0146 \beta \left[\frac{2\sigma}{g(\rho_l - \rho_v)} \right]^{1/2}$$

where C_1 is the empirical constant and P_r refers to the reduced pressure of saturated ammonia ($P_r = P_{sat}/P_c$). The constant C_1 and exponent $n(P_r)$ were determined to be best fitted with the measured data, and the new correlation for the case of decreasing heat flux was suggested in the following Eq. (12).

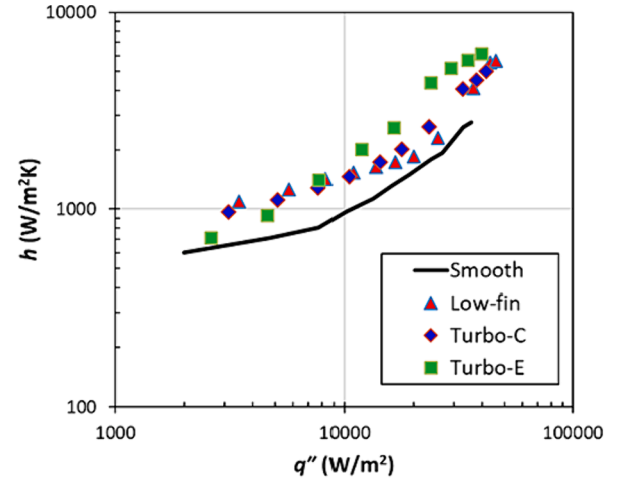
$$\frac{h_b D_b}{k_l} = (193.42) \left(\frac{q'' D_b}{k_l T_{sat}} \right)^{1.778 P_r + 0.376} \left(\frac{\rho_v}{\rho_l} \right)^{0.581} (Pr_l)^{0.533} \quad (12)$$

Fig. 6 shows the comparison between the measured HTC's and those calculated from Eq. (12) in the case of decreasing heat flux for the data of the present study as well as the Djundin [18]'s data. At Djundin [18]'s measurement condition (the saturation pressure of 8.57 bar and the heat flux range from 2100 to 24,700 W/m²), the suggested correlation shows acceptable prediction (error ~ 7.61 %) for the heat transfer coefficient. The mean absolute error (MAE) of all the calculated HTC's including the present data and Djundin's data was computed as 4.22 %.

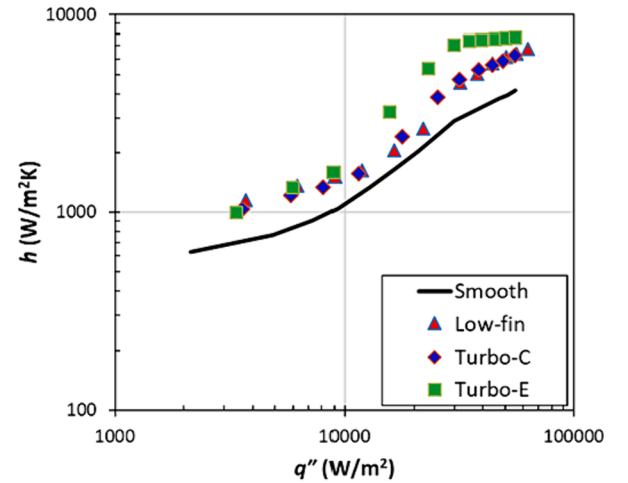
3.2. Boiling heat transfer enhancement on the enhanced tubes

Fig. 7 shows boiling HTC's on three enhanced tubes of Low-fin, Turbo-C, and Turbo-E compared with those on the smooth tube when the heat flux decreases. All three enhanced tubes exhibit considerable enhancement in boiling heat transfer over the smooth tube at three different saturation pressures of 6.15, 8.57, and 11.7 bar. The Low-fin tube has numerous fins with a height of 1.33 mm on the outer surface of the tube, which are effective in augmenting both the heat transfer area and bubble nucleation sites. Because the HTC is calculated based on the tube outer diameter, D_o , the augmentation in both heat transfer area and nucleation sites owing to the low fins is directly related to the enhancement of the HTC for the Low-fin tube. The Turbo-C tube shows somewhat lower HTC's than the Low-fin tube although both tubes have similar fin geometries. The Turbo-C tube also has low fins on the outer surface with a height of 1.205 mm (slightly lower than the Low-fin tube), and it has numerous oblique cuttings in the middle of low fins. The oblique cuttings do not help improve the boiling heat transfer performance and rather seem to deteriorate the boiling performance by reducing the heat transfer area. Near the heat flux of 10,000 W/m² at the saturation pressure of 8.57 bar, the HTC's for the smooth tube and three enhanced tubes (Low-fin, Turbo-C, and Turbo-E) are 9767 W/m² K ($q'' = 9767$ W/m²), 3829 W/m² K ($q'' = 10,384$ W/m²), 2987 W/m² K ($q'' = 9767$ W/m²), and 4415 W/m² K ($q'' = 10,750$ W/m²), respectively. Three enhanced tubes of Low-fin, Turbo-C, and Turbo-E exhibit a considerable increase in the HTC's over the smooth surface by 150 %, 95 %, and 188 %, respectively. The large enhancement in boiling HTC's on the enhanced tubes primarily results from an increase in the heat transfer area through fins.

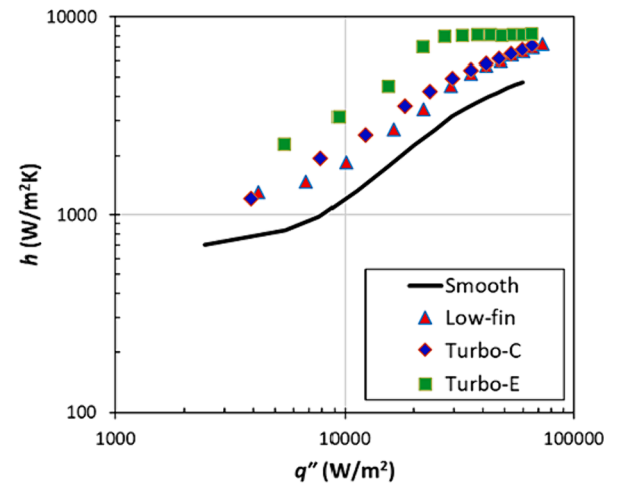
It seems that the Turbo-E tube has additional enhancement in the heat transfer from an increase in bubble nucleation sites through re-entrant type cavities on the tube surface. The improvement of the boiling performance using the Turbo-E tube is prominent at the high saturation pressures of 8.57 and 11.7 bar. In contrast, at the low saturation pressure of 6.15 bar, the HTC's for the Turbo-E tube are inferior to those for the Low-fin tube, except for the high heat flux region above 30,000 W/m². Near the heat flux of 10,000 W/m², the HTC's for three enhanced tubes (Low-fin, Turbo-C, and Turbo-E) are 3360 W/m² K ($q'' = 9093$ W/m²), 2836 W/m² K ($q'' = 11,626$ W/m²), and 3069 W/m² K ($q'' = 11,531$ W/m²), respectively. Approximately 9 % higher HTC is obtained for the Low-fin tube compared with the Turbo-E tube. In comparison, even at the same saturation pressure of 6.15 bar, in case of



(a) $P_{sat} = 6.15$ bar ($T_{sat} = 10$ °C)



(b) $P_{sat} = 8.57$ bar ($T_{sat} = 20$ °C)



(c) $P_{sat} = 11.7$ bar ($T_{sat} = 30$ °C)

Fig. 8. Boiling HTC's on three enhanced tubes of Low-fin, Turbo-C, and Turbo-E compared with those on the smooth tube (in case of increasing the heat flux).

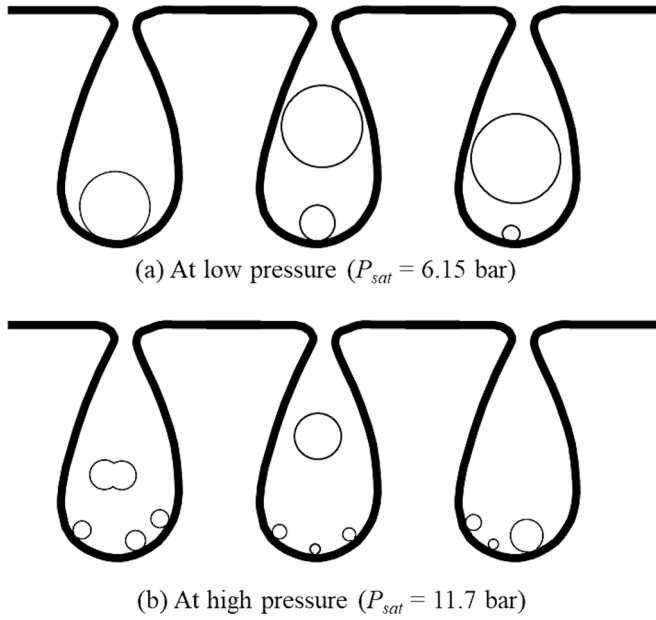


Fig. 9. Schematic drawings of bubble generation, growth, and departure inside the cavity on the Turbo-E tube at low ($P_{sat} = 6.15$ bar) and high ($P_{sat} = 11.7$ bar) saturation pressures.

increasing the heat flux, the Turbo-E tube certainly shows higher HTC's than the Low-fin tube as shown in Fig. 8. The superior performance of the Turbo-E tube confirms that the re-entrant type cavities on the Turbo-E tube facilitates earlier bubble nucleation in the cavity than the Low-fin tube. However, once the nucleation sites are fully activated, heat transfer performance for the Low-fin tube exceeds that for the Turbo-E tube in case of decreasing the heat flux at the saturation pressure of 6.15 bar. The difference in superiority between Low-fin and Turbo-E according to the saturation pressure is ascribed to the boiling HTC's for the Turbo-E tube are highly affected by the saturation pressure.

For the Turbo-E tube, the bubble nucleation and the bubble growth occur inside the re-entrant cavity, which is followed by the bubble departure. The behaviors of the bubble generation, growth, and departure at two saturation pressures of 6.15 and 11.7 bar are schematically

depicted in Fig. 9. At the low saturation pressure of 6.15 bar, bubble departure diameter (D_b) at the saturation temperature is calculated to be 0.438 mm from the Cole and Rohsenow's correlation as shown below [32].

$$Bo_D^{1/2} = C(Ja^*)^{5/4} \quad (13)$$

where $C = 4.65 \times 10^{-4}$ for fluids other than water

$$Bo_D = \frac{g(\rho_l - \rho_v)D_b^2}{\sigma}$$

$$Ja^* = \frac{T_{sat}c_{pl}\rho_l}{\rho_v h_{fg}}$$

The bubble departure diameter at the low saturation pressure is comparable to the cavity characteristic length (cavity length ~ 0.770 mm and cavity width ~ 0.280 mm) of the Turbo-E tube as presented in the cross-section of Fig. 3(c). As the bubble is large enough to block the flow passage in the cavity, convective liquid flow is disturbed, and the heat transfer in the cavity is dominated by the latent heat transport. In contrast, the bubble departure diameter at the high saturation pressure of 11.7 bar is calculated to be 0.217 mm using the identical correlation (Eq. (13)). The diameter is merely 49.5 % of that at the saturation pressure of 6.15 bar and the bubble departure volume is reduced to 12.2 %. In this case, multiple bubbles can be generated inside the cavity because the wall superheat required to activate the wall nucleation sites decreases with the saturation pressure. Small bubbles depart from the wall and agglomerate into a larger bubble, but it is still smaller than that at the low saturation pressure. Small bubbles are advantageous in generating convective liquid flow inside the cavity, which is beneficial to increase heat transfer between the heated wall and liquid ammonia. The development in liquid convection inside the cavity is responsible for the large improvement of the HTC's for the Turbo-E tube at high saturation pressure. On the contrary, the large bubble at the low saturation pressure hinders the liquid convection, resulting in the deterioration in heat transfer inside the cavity.

Fig. 10 shows captured images of nucleate boiling phenomena on the smooth tube and three enhanced tubes near the heat flux of 30,000 W/m² at the saturation pressure of 8.57 bar ($T_{sat} = 20$ °C). The Low-fin tube exhibits similar bubble nucleation behavior compared to the smooth tube, but the number of generated bubbles on the Low-fin tube is slightly

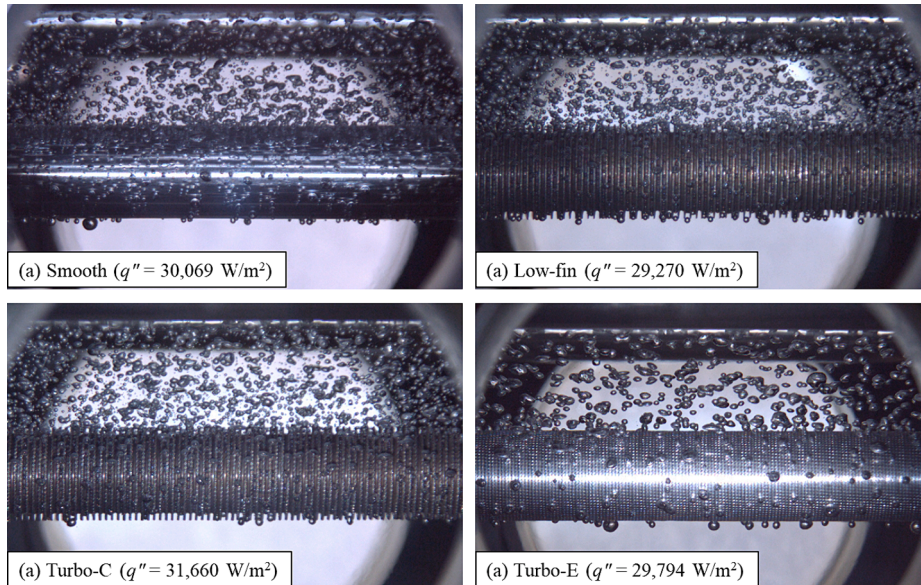


Fig. 10. Captured images of nucleate boiling phenomena on smooth and three enhanced tubes near the heat flux of 30,000 W/m² at the saturation pressure of 8.57 bar ($T_{sat} = 20$ °C).

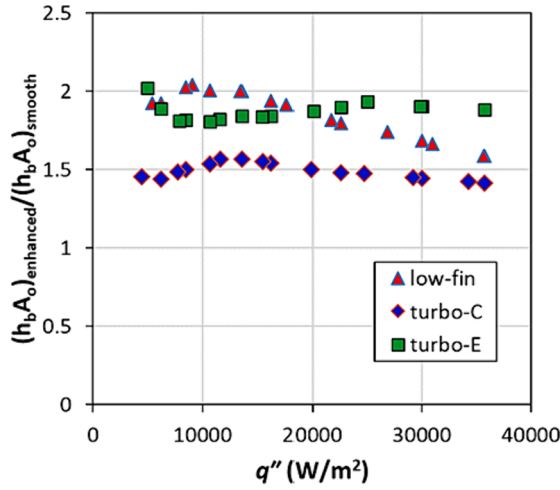
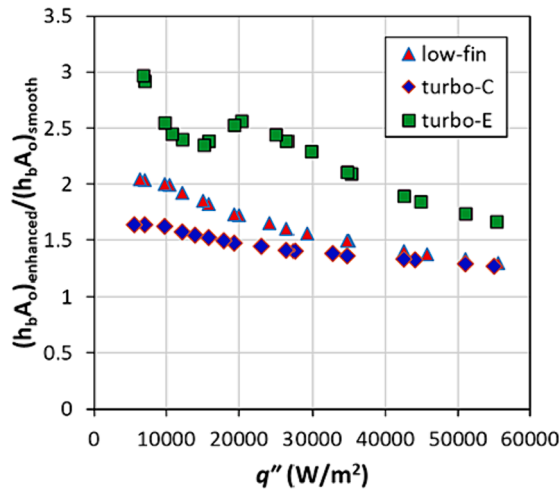
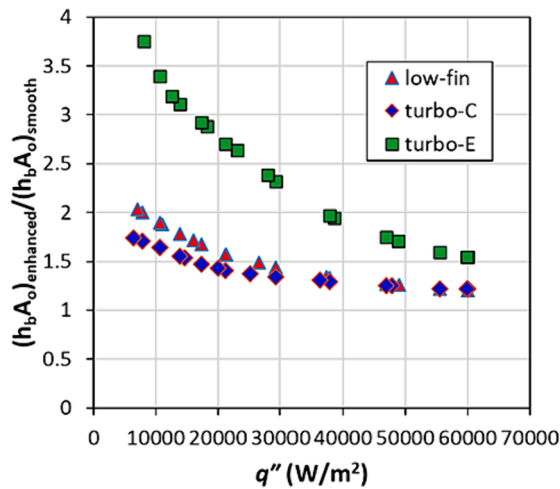
(a) $P_{\text{sat}} = 6.15 \text{ bar}$ ($T_{\text{sat}} = 10 \text{ }^{\circ}\text{C}$)(b) $P_{\text{sat}} = 8.57 \text{ bar}$ ($T_{\text{sat}} = 20 \text{ }^{\circ}\text{C}$)(c) $P_{\text{sat}} = 11.7 \text{ bar}$ ($T_{\text{sat}} = 30 \text{ }^{\circ}\text{C}$)

Fig. 11. The ratios of thermal conductance of three enhanced tubes to the smooth tube according to the saturation pressure from 6.15 to 11.7 bar (when decreasing the heat flux).

Table 2

Summary of average enhancement factors and overall fin efficiency for three enhanced tubes.

	P_{sat} (bar)	average enhancement factor			overall fin efficiency		
		Low-fin	Turbo-C	Turbo-E	Low-fin	Turbo-C	Turbo-E
decreasing q''	6.15	1.85	1.49	1.87	0.41	0.52	0.45
	8.57	1.68	1.45	2.31	0.44	0.53	0.47
	11.7	1.57	1.42	2.43	0.45	0.58	0.55
increasing q''	6.15	1.18	1.26	1.75	0.68	0.63	0.49
	8.57	1.21	1.23	1.81	0.62	0.63	0.61
	11.7	1.22	1.33	2.09	0.59	0.63	0.72

higher than the smooth tube. Because bubble nucleation behavior looks similar at both smooth and Low-fin tubes, the HTC on those tubes are highly dependent upon the heat transfer area. Since the Low-fin tube has a larger heat transfer area owing to fins, it shows higher boiling heat transfer performance than the smooth tube. In comparison, The Turbo-E tube exhibits much different bubble nucleation behavior from both smooth and Low-fin tubes. It is seen in Fig. 10 that the bubble departure diameter for the Turbo-E tube is much larger than smooth and other enhanced tubes. As already presented in the schematic drawings in Fig. 9, for the Turbo-E tube, small bubbles are generated inside the re-entrant cavity and agglomerated into a larger bubble. The agglomerated bubble passes through the narrow cavity mouth and releases out to the ammonia liquid pool. Even though the number of apparently departing bubbles is smaller than the smooth and other enhanced tubes, the actual number of activated nucleation sites inside the re-entrant cavity is presumed to exceed those on other tubes. As a result, the Turbo-E tube exhibits the highest HTCs among smooth and enhanced tubes, except for the case of decreasing the heat flux at the low saturation pressure of 6.15 bar.

3.3. Heat transfer enhancement factors on the enhanced tubes

Thermal performance for three enhanced tubes, Low-fin, Turbo-C, and Turbo-E, can be compared with the smooth tube using the HTC ratio as shown in the following equation.

$$HTC_{\text{ratio}} = \frac{(h_b)_{\text{enhanced}}}{(h_b)_{\text{smooth}}} \quad (14)$$

However, because each tube has a different tube outer diameter, D_o , the HTC ratio between the smooth tube and enhanced tube does not reflect the ratio of actual thermal performance. The ratio of thermal conductance, $h_b A_o$, as shown in Eq. (15) is more appropriate for the thermal performance comparison between the smooth tube and the enhanced tube.

$$\text{Thermal conductance ratio} = \frac{(h_b A_o)_{\text{enhanced}}}{(h_b A_o)_{\text{smooth}}} \quad (15)$$

Fig. 11 shows the ratios of thermal conductance between the smooth tube and three enhanced tubes according to the saturation pressure from 6.15 to 11.7 bar when decreasing the heat flux. As already discussed in the previous section, the Turbo-E tube shows a large enhancement in the thermal conductance over the smooth tube at the high saturation pressures of 8.57 and 11.7 bar. The enhancement factor of thermal conductance is as high as 3.76, especially at the low heat flux region below 10,000 W/m², because of the earlier bubble nucleation in the re-entrant cavities. In contrast, at the low saturation pressure of 6.15 bar ($T_{\text{sat}} = 10 \text{ }^{\circ}\text{C}$), the Turbo-E tube exhibits the highest enhancement of thermal conductance only at the heat flux above 20,000 W/m². At the low heat flux region, the Low-fin tube shows a higher enhancement factor than the Turbo-E tube. As a result, the average enhancement

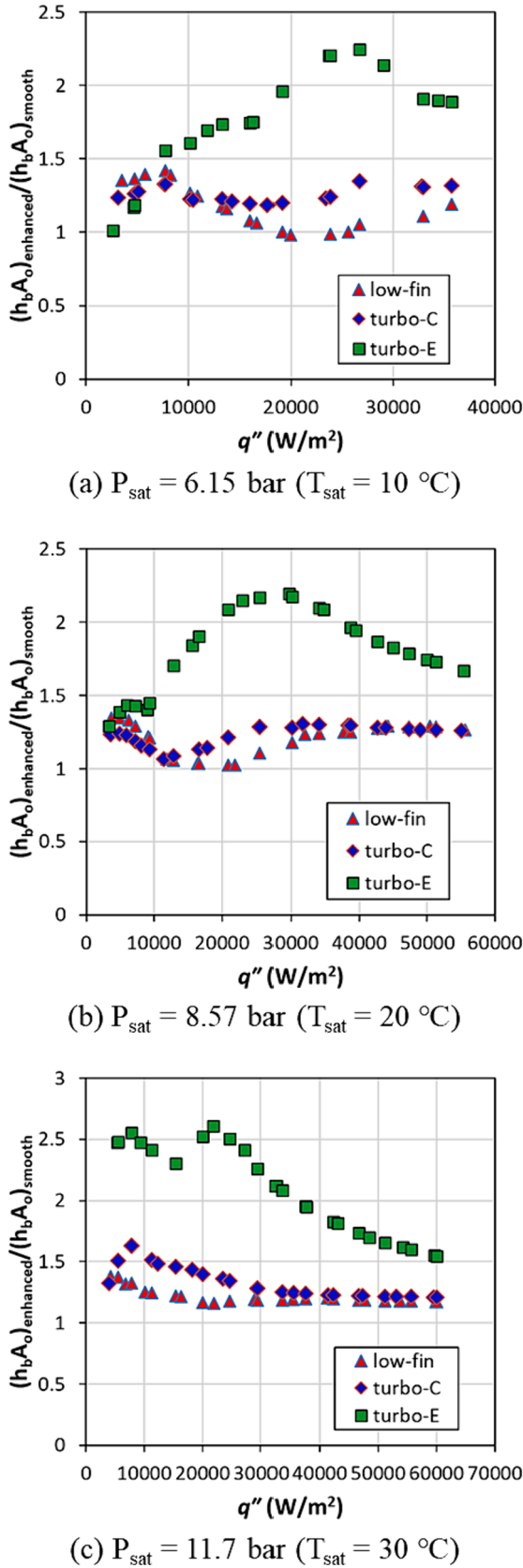


Fig. 12. The ratios of thermal conductance of three enhanced tubes to the smooth tube according to the saturation pressure from 6.15 to 11.7 bar (when increasing the heat flux).

factors for the Low-fin tube and Turbo-E tube are similar as 1.85 and 1.87, respectively, at the saturation pressure of 6.15 bar. Table 2 shows the summary of the enhancement factors of thermal conductance for three enhanced tubes in both cases of decreasing the heat flux and increasing the heat flux. When decreasing the heat flux, the Turbo-E tube shows the highest enhancement, the Low-fin tube is the next, and the Turbo-C tube is the last. The average enhancement factor in case of decreasing the heat flux ranges from 1.42 (Turbo-C) to 2.43 (Turbo-E).

Fig. 12 displays the ratios of thermal conductance between a smooth tube and three enhanced tubes when increasing the heat flux. At all saturation pressures, the Turbo-E tube shows larger enhancement factors than both the Low-fin tube and Turbo-C tube because of the earlier bubble nucleation inside the re-entrant cavity. While the enhancement factors of the Low-fin tube are higher than those of the Turbo-C tube when decreasing the heat flux, when increasing the heat flux, the Turbo-C tube exhibits better enhancement factors. The better performance of the Turbo-C tube is ascribed to the oblique cuttings on the Turbo-C tube because the additional cuttings bring about an increase in promising nucleation sites. But, once the nucleate boiling begins, the Low-fin tube has more benefit from the larger finned surface area compared to the Turbo-C tube when decreasing the heat flux. From the summary in Table 2, average enhancement factors when increasing the heat flux are relatively small compared to those when decreasing the heat flux. The average enhancement factor in case of increasing the heat flux ranges from 1.18 (Low-fin) to 2.09 (Turbo-E).

The overall fin efficiency, η_o , on the enhanced tubes was obtained from the following relations.

$$Q_{\text{enhanced tube}} = \eta_o (h_b A_o)_{\text{smooth}} A_t \Delta T_w = \eta_o (h_b A_o)_{\text{smooth}} \frac{A_t}{A_{o,\text{smooth}}} \Delta T_w \quad (16)$$

where A_t refers to the total surface area including both base and finned area. The heat transfer rate on the enhanced tube can also be calculated based on the heat transfer coefficient on the enhanced tube.

$$Q_{\text{enhanced tube}} = (h_b A_o)_{\text{enhanced}} \Delta T_w \quad (17)$$

Combining Eq. (16) and (17),

$$\eta_o = \frac{(h_b A_o)_{\text{enhanced}}}{(h_b A_o)_{\text{smooth}}} \left(\frac{A_{o,\text{smooth}}}{A_t} \right) = (\text{enhancement factor}) \left(\frac{A_{o,\text{smooth}}}{A_t} \right) \quad (18)$$

The last term in the left side is the inverse of the area enhancement ratio of enhanced tube to the smooth tube as summarized in Table 1. As a result, the overall fin efficiency can be obtained by multiplying the heat transfer enhancement factor and the area enhancement ratio of the enhanced tube. As presented in Table 2, the overall fin efficiency averaged over the entire enhanced tubes tested is as low as 0.56. The low fin efficiency on the enhanced tubes is ascribed to that the bubble nucleation is not increased as much as the increase in the surface area, especially for the low finned surfaces.

The boiling HTC on the enhanced tubes can be obtained from the following procedure. First, the boiling HTC on the smooth tube is acquired from either the previous Stephan-Abdelsalam correlation in case of increasing the heat flux or our proposed correlation, Eq. (12) in case of decreasing the heat flux. Then, the average HTCs on the enhanced tubes are calculated from the relevant enhancement factor summarized in Table 2 according to the saturation pressure and whether heat flux is increasing or decreasing. The individual HTC at each heat flux can be obtained from the enhancement factors graphically presented in Figs. 11 and 12.

4. Conclusions

Pool boiling heat transfer of ammonia was investigated on a smooth tube and three enhanced tubes at the saturation pressure of 6.15, 8.57, and 11.7 bar. The following conclusions were drawn from the measured

data and visual observation.

- Boiling hysteresis of ammonia on a smooth tube was observed between when the heat flux increases and when the heat flux decreases, which resulted in the disparity of the HTC's from the previous correlations when the heat flux decreases. A suitable correlation modified from the Stephan-Abdelsalam was suggested to compensate for the deviation of the HTC's in case of decreasing the heat flux.
- All three enhanced tubes of Low-fin, Turbo-C, and Turbo-E showed considerable enhancement in the HTC's compared with the smooth tube. Turbo-E tube with re-entrant cavities exhibited the best enhancement at the high saturation pressures of 8.57 and 11.7 bar owing to earlier bubble nucleation inside the cavity, however, its performance was degraded at the low saturation pressure of 6.15 bar because a larger bubble disturbs the convective liquid flow inside the cavity.
- The boiling heat transfer performance of three enhanced tubes was summarized by the enhancement factor defined as the ratio of thermal conductance of the enhanced tube over the smooth tube. The average enhancement factor ranges from 1.42 to 2.43 in case of decreasing the heat flux and from 1.18 to 2.09 in case of increasing the heat flux.
- The boiling heat transfer performance on the enhanced tube can be calculated by combining the thermal conductance obtained from the smooth tube correlation and the enhancement factor on the enhanced tube.

CRediT authorship contribution statement

Jin Sub Kim: Investigation, Writing – original draft. **Wookyoung Kim:** Data curation, Methodology. **Hak Soo Kim:** Validation, Writing – review & editing. **Young Kim:** Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

This work was supported by Korea Institute of Energy Technology Evaluation and Planning (KETEP) and the Ministry of Trade, Industry & Energy (MOTIE) of the Republic of Korea. (Grant No. 20212050100010).

References

- [1] IEA, Net Zero by 2050 : A Roadmap for the Global Energy Sector, (2021).
- [2] O. Zurcher, J.R. Thome, D. Favrat, Evaporation of ammonia in a smooth horizontal tube: heat transfer measurements and predictions, *J. Heat Transfer*. 121 (1999) 89–101, <https://doi.org/10.1115/1.2825974>.
- [3] H. Arima, J.H. Kim, A. Okamoto, Y. Ikegami, Local boiling heat transfer characteristics of ammonia in a vertical plate evaporator, *Int. J. Refrig.* 33 (2010) 359–370, <https://doi.org/10.1016/j.ijrefrig.2009.09.017>.
- [4] M.H. Maqbool, B. Palm, R. Khodabandeh, Boiling heat transfer of ammonia in vertical smooth mini channels: experimental results and predictions, *Int. J. Therm. Sci.* 54 (2012) 13–21, <https://doi.org/10.1016/j.jthermalsci.2011.09.012>.
- [5] X. Zeng, M.C. Chyu, Z.H. Ayub, Experimental investigation of ammonia spray evaporator with triangular-pitch plain tube bundle, part II: evaporator performance, *Int. J. Heat Mass Transf.* 44 (2001) 2081–2092, [https://doi.org/10.1016/S0017-9310\(00\)00251-9](https://doi.org/10.1016/S0017-9310(00)00251-9).
- [6] B.H. Yang, H. Wang, X. Zhu, Q. Liao, Y.D. Ding, R. Chen, Heat transfer enhancement of spray cooling with ammonia by microcavity surfaces, *Appl. Therm. Eng.* 50 (2013) 245–250, <https://doi.org/10.1016/j.applthermaleng.2012.06.029>.
- [7] H. Wang, J.J. Wu, Q. Yang, X. Zhu, Q. Liao, Heat transfer enhancement of ammonia spray cooling by surface modification, *Int. J. Heat Mass Transf.* 101 (2016) 60–68, <https://doi.org/10.1016/j.ijheatmasstransfer.2016.05.052>.
- [8] T. Inoue, M. Monde, Y. Teruya, Pool boiling heat transfer in binary mixtures of ammonia/water, *Int. J. Heat Mass Transf.* 45 (2002) 4409–4415, [https://doi.org/10.1016/S0017-9310\(02\)00153-9](https://doi.org/10.1016/S0017-9310(02)00153-9).
- [9] H. Arima, M. Monde, Y. Mitsutake, Heat transfer in pool boiling of ammonia/water mixture, *Heat Mass Transf. Und Stoffuebertragung*. 39 (2003) 535–543, <https://doi.org/10.1007/s00231-002-0302-2>.
- [10] J.X. Zheng, G.P. Jin, M.C. Chyu, Z.H. Ayub, Flooded boiling of ammonia with miscible oil outside a horizontal plain tube, *HVAC R Res.* 7 (2001) 185–204, <https://doi.org/10.1080/10789669.2001.10391270>.
- [11] A. Abbas, Z.H. Ayub, T.S. Khan, A.H. Ayub, J.A. Chattha, A review of correlations for outside boiling of ammonia on single tube and bundles, *Heat Transf. Eng.* 39 (2018) 1425–1436, <https://doi.org/10.1080/01457632.2017.1379335>.
- [12] M.C. Chyu, J. Zheng, Z. Ayub, Bundle effect of ammonia/lubricant mixture boiling on a horizontal bundle with enhanced tubing and inlet quality, *Int. J. Refrig.* 32 (2009) 1876–1885, <https://doi.org/10.1016/j.ijrefrig.2009.07.013>.
- [13] J. Fernández-Seara, Á.A. Pardiñas, R. Diz, Experimental heat transfer coefficients of pool boiling and spray evaporation of ammonia on a horizontal plain tube, *Int. J. Refrig.* 67 (2016) 259–270, <https://doi.org/10.1016/j.ijrefrig.2016.02.019>.
- [14] J. Fernández-Seara, Á. Pardiñas, R. Diz, Heat transfer enhancement of ammonia pool boiling with an integral-fin tube, *Int. J. Refrig.* 69 (2016) 175–185, <https://doi.org/10.1016/j.ijrefrig.2016.05.012>.
- [15] K. Spindler, Overview and discussion on pool boiling heat transfer data and correlations of ammonia, *Int. J. Refrig.* 33 (2010) 1292–1306, <https://doi.org/10.1016/j.ijrefrig.2010.06.012>.
- [16] Z. Ayub, Current and future prospects of enhanced heat transfer in ammonia systems, *Int. J. Refrig.* 31 (2008) 652–657, <https://doi.org/10.1016/j.ijrefrig.2007.11.012>.
- [17] G.N. Danilova, Effect of surface conditions on boiling heat transfer of refrigerants in shell-and-tube evaporators, *Heat Transf. - Sov. Res.* 22 (1990) 56–65.
- [18] V.A. Djundin, Influence of the type of surface on heat transfer in boiling, *Kholod. Tekh.* 5 (1984) 33–37.
- [19] S. Kotthoff, D. Gorenflo, Pool boiling heat transfer to hydrocarbons and ammonia: a state-of-the-art review, *Int. J. Refrig.* 31 (2008) 573–602, <https://doi.org/10.1016/j.ijrefrig.2008.03.008>.
- [20] X. Zeng, M.C. Chyu, Z.H. Ayub, Evaporation heat transfer performance of nozzle-sprayed ammonia on a horizontal tube, *ASHRAE Trans.* 2 (1995) 136–149, <https://doi.org/10.1615/ichmt.2000.thersieprocvol2thersieprocvol1.420>.
- [21] A. Kuprianova, Heat transfer with pool boiling of ammonia on horizontal tubes, *Kholodiln. Tekh.* 47 (1970) 40–44.
- [22] I.L. Mostinski, Application of the rule of corresponding states for calculation of heat transfer and critical heat flux, *Teploenergetika*. 4 (1963) 66–71.
- [23] D. Gorenflo, E. Baumhögger, G. Herres, S. Kotthoff, Prediction methods for pool boiling heat transfer: a state-of-the-art review, *Int. J. Refrig.* 43 (2014) 203–226, <https://doi.org/10.1016/j.ijrefrig.2013.12.012>.
- [24] B.B. Mikic, W.M. Rohsenow, A new correlation of pool-boiling data including the effect of heating surface characteristics, *J. Heat Transfer*. 91 (1969) 245–250, <https://doi.org/10.1115/1.3580136>.
- [25] K. Stephan, M. Abdelsalam, Heat-transfer correlations for natural convection boiling, *Int. J. Heat Mass Transf.* 23 (1980) 73–87, [https://doi.org/10.1016/0017-9310\(80\)90140-4](https://doi.org/10.1016/0017-9310(80)90140-4).
- [26] V.P. Carey, Liquid-vapor phase-change phenomena: an introduction to the thermophysics of vaporization and condensation processes in heat transfer equipment, CRC Press, 2020.
- [27] J.S. Kim, W. Kim, H.S. Kim, Y. Kim, S.H. Yoon, Pool boiling heat transfer of ammonia outside a single tube with fin structures: hysteresis phenomena and boiling enhancement, *Int. Commun. Heat Mass Transf.* 149 (2023) 107157.
- [28] V. Gnielinski, New equations for heat and mass transfer in turbulent pipe and channel flow, *Int. Chem. Eng.* 16 (1976) 359–367.
- [29] R.J. Moffat, Describing the uncertainties in experimental results, *Exp. Therm. Fluid Sci.* 1 (1988) 3–17, [https://doi.org/10.1016/0894-1777\(88\)90043-X](https://doi.org/10.1016/0894-1777(88)90043-X).
- [30] D. Taler, J. Taler, Simple heat transfer correlations for turbulent tube flow, in: *IN E3S Web of Conferences* 13 EDP Sciences, 2017, p. 02008.
- [31] M. Shahidi, M.R. Aligoodarzi, M.A. Akhavan-Behabadi, S. Foroutani, A. Rahbari, Experimental and numerical investigation on turbulent flow of MWCNT-water nanofluid inside vertical coiled wire inserted tubes, *Thermal Science* (2016).
- [32] R. Cole, W.M. Rohsenow, Correlation of bubble departure diameters for boiling of saturated liquids, in: *Chem. Eng. Prog. Symp. Ser.* 65 (1969) 211–213.